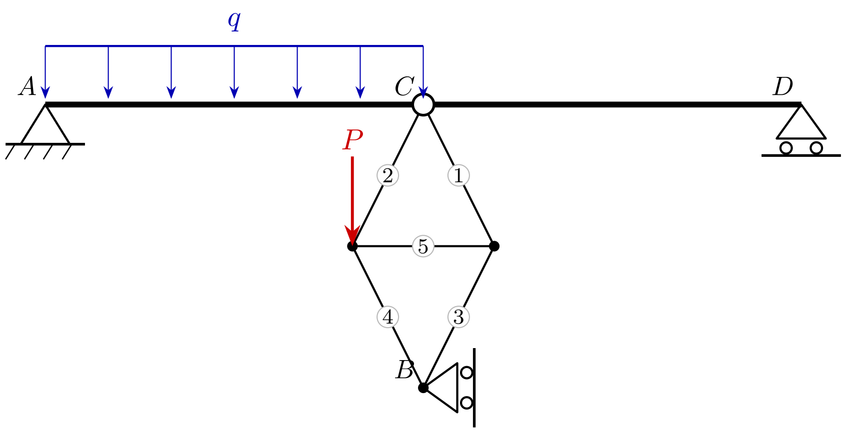
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k, reactions a, interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \;\Rightarrow\; \text{statically determinate}$$

2. Support reactions and hinge forces

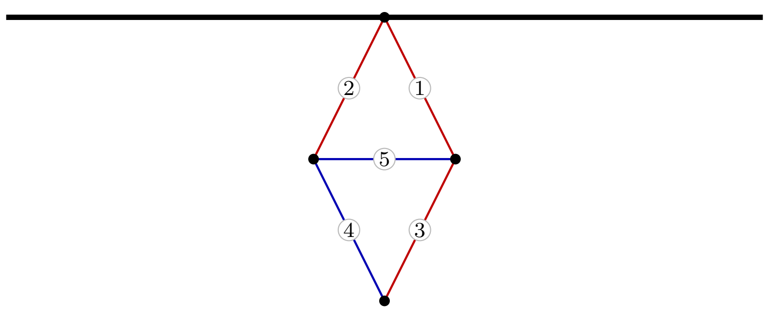
$A:\; R_x = 3,\; R_y = 21 \quad D:\; R_x = 0,\; R_y = 11 \quad B:\; R_x = -3,\; R_y = 0$

$C:\; H_x = -3,\; H_y = -12$

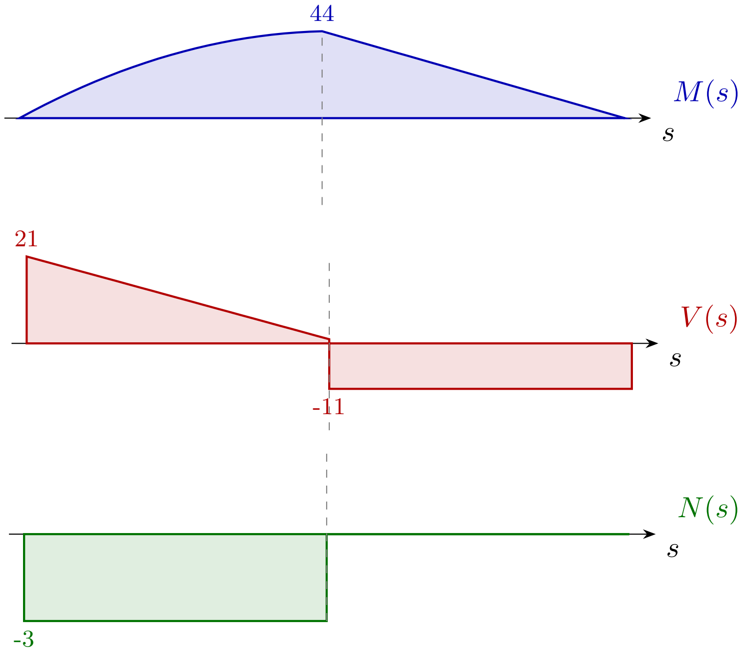
3. Truss member forces

Member	Force S [kN]	Type
1	3.35	tension
2	10.06	tension
3	3.35	tension
4	-3.35	compression
5	-3	compression

red: tension (+) blue: compression (−)



4. Section-force diagrams of the beams


44
$M(s)$
s
21
$V(s)$
s
-11
$N(s)$
s
-3